SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Fundamentals Of Data Science **COURSE CODE: DSA0402**

COURSE OUTCOME COVERED

CO3: *Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 10 **Total Marks:100 marks**

Annexure A

POSTER PRESENTATION

Code of Conduct:

*I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jannathul Firdaus. N

RUBRICS FOR CALCULATION

Poster Presentation							
Criteria	**Weightage (%)**	**Level 4 (Exemplary) 5 Marks**	**Level 3 (Proficient) 4 Marks**	**Level 2 (Developing) 3 – 2 Marks**	**Level 1 (Beginning) 1 Marks**	**Level**	**Marks**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood		
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach		
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals		
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides		
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness		

Annexure A

Question 01: University Learning Management System (LMS) Usage Analysis (100 Marks)

A university uses an online Learning Management System (LMS) to deliver course materials. Most students spend **2–4 hours per week** on the platform, while a small number of students spend **more than 20 hours per week**. The university wants to understand student engagement patterns and identify factors affecting academic performance.

Prepare a comprehensive poster presentation that includes:

1. Data summarization of LMS usage statistics.
2. Analysis of the distribution of student engagement data.
3. Identification and interpretation of outliers.
4. Discussion on whether highly active students should be considered outliers.
5. Analysis of skewness and kurtosis in LMS usage patterns.
6. Relationship between LMS usage and academic performance.
7. Recommendations for improving student engagement and learning outcomes.

Question 02: Renewable Energy Production Analysis (100 Marks)

A solar power plant generates electricity throughout the year. On normal days, energy production ranges from **5,000 to 8,000 kWh**, while on exceptionally sunny days production exceeds **12,000 kWh**. During cloudy or rainy days, production may fall below **2,000 kWh**. The management wants to analyze production variability and improve energy forecasting.

Develop a poster presentation covering the following aspects:

1. Summarize the energy production data using descriptive statistics.
2. Examine the distribution of daily energy generation.
3. Identify and analyze unusually high and low production values.
4. Discuss appropriate methods for handling outliers.
5. Measure and interpret skewness and kurtosis.
6. Evaluate the impact of weather conditions on data distribution.
7. Suggest forecasting and operational strategies to improve renewable energy management.

Question 03: E-Commerce Festival Sales Analysis (100 Marks)

An online retailer records average daily sales of ₹10 lakh. During a festival season, sales increase to ₹2 crore per day.

Develop a poster discussing:

1. Summary statistics before and during the festival.
2. Changes in sales distribution.
3. Identification of unusual sales patterns.
4. Whether festival sales should be treated as outliers.
5. Interpretation of skewness and kurtosis.

6. Business insights obtained from the analysis.
7. Strategic recommendations for inventory planning.

Question 04: Healthcare Monitoring System (100 Marks)

A hospital continuously monitors patient heart rates using wearable sensors. Most readings are between 60–90 bpm, while some readings exceed 180 bpm.

Prepare a poster that addresses:

1. Data summarization of patient health records.
2. Visualization techniques for monitoring trends.
3. Detection of abnormal readings.
4. Methods for treating erroneous sensor values.
5. Analysis of skewness and kurtosis.
6. Differentiating medical emergencies from data errors.
7. Recommendations for real-time healthcare monitoring.

Question 05: Cybersecurity Network Traffic Analysis (100 Marks)

A company's server typically receives 20,000 requests per hour. During a cyberattack, the traffic suddenly increases to 2 million requests per hour.

Prepare a comprehensive poster explaining:

1. Methods for summarizing network traffic data.
2. Traffic distribution under normal conditions.
3. Detection of abnormal traffic spikes.
4. Outlier treatment strategies.
5. Analysis of skewness and kurtosis.
6. Role of anomaly detection in cybersecurity.
7. Security measures to prevent future attacks.

Question 06: Manufacturing Quality Control (100 Marks)

A manufacturing company produces 50,000 bottles daily. The target volume is 1 liter, but some bottles contain less than 900 ml or more than 1.1 liters.

Design a poster discussing:

1. Data summarization techniques.
2. Distribution of bottle volumes.
3. Detection of defective products using outlier analysis.
4. Treatment of abnormal measurements.
5. Analysis of skewness and kurtosis.
6. Impact on product quality and customer satisfaction.
7. Recommendations for process improvement.

Question 07: Climate Change and Weather Analysis (100 Marks)

A city normally experiences temperatures between 25°C and 35°C. Over the last decade, several days recorded temperatures above 45°C.

Prepare a poster presentation covering:

1. Summary statistics of temperature data.
2. Distribution of climatic variables.
3. Identification of extreme weather events.
4. Outlier treatment considerations.
5. Analysis of skewness and kurtosis.
6. Implications for climate change research.
7. Recommendations for policymakers.

Question 08: Social Media Engagement Analysis (100 Marks)

A content creator uploads 500 videos. Most videos receive fewer than 10,000 views, while a few viral videos receive over 10 million views.

Develop a poster explaining:

1. Data summarization methods.
2. Distribution of engagement metrics.
3. Detection of viral-content outliers.
4. Whether viral videos should be retained.
5. Interpretation of skewness and kurtosis.
6. Business implications for content creators.
7. Strategies to improve audience engagement.

Question 09: Stock Market Volatility Analysis (100 Marks)

A stock usually fluctuates within ±2% daily. Following a major corporate announcement, the stock price rises by 30% in a single day.

Prepare a detailed poster discussing:

1. Summary statistics of stock returns.
2. Distribution of financial data.
3. Detection of abnormal market movements.
4. Treatment of extreme observations.
5. Interpretation of skewness.
6. Importance of kurtosis in financial risk analysis.
7. Investment recommendations based on findings.

Question 10: Smart City Traffic Management System (100 Marks)

A smart city uses sensors to monitor traffic flow. The average traffic volume is 700 vehicles per hour. During a major cultural festival, traffic increases to 7,000 vehicles per hour.

Create a poster presentation that includes:

1. Data summarization of traffic patterns.
2. Analysis of traffic distribution.
3. Identification of unusual traffic events.
4. Outlier detection and treatment methods.
5. Analysis of skewness and kurtosis.
6. Traffic management recommendations.
7. Development of a predictive model for future events.

Question 01: University LMS Usage Analysis

Understanding Student Engagement & Academic Performance

1. Data Summarization Methods

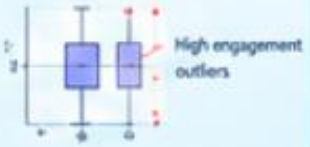
- Mean, Median, Mode
- Quartiles & IQR
- Percentage distribution
- Cross-tabulation

2. Distribution of LMS Usage



3. Outlier Detection

- Students spending >20 hours are potential outliers.
- Use box plot/IQR method.



4. Factors Affecting Performance

- LMS usage hours
- Access to resources
- Study habits
- Demographics (year, program)

5. Insights & Implications

- Most students spend 2–4 hours/week.
- High engagement (>20 hrs) is rare.
- Engagement positively correlates with better academic performance.

6. Recommendations

- Increase interactive content.
- Provide personalized support.
- Promote consistent usage and habit building.

More Engagement → Better Learning → Higher Performance

Renewable Energy Production Analysis

Analyzing Production Variability & Improving Forecasting



1. Data Summarization

Summarize energy production data using descriptive statistics.

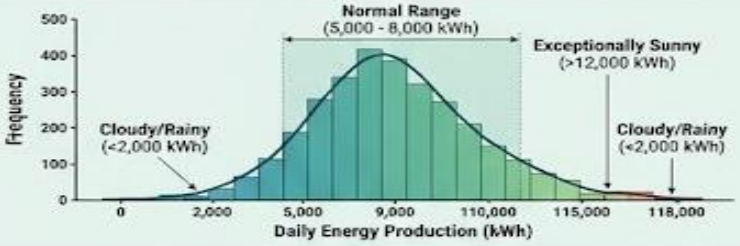
Key Metrics	
Total Generation	2.5 MWh
Mean Daily Production	6,800 kWh
Median Production	6,500 kWh
Standard Deviation	2,100 kWh





2. Energy Distribution

Examine the distribution of daily energy generation.

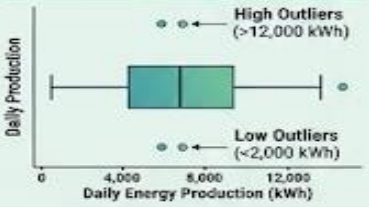




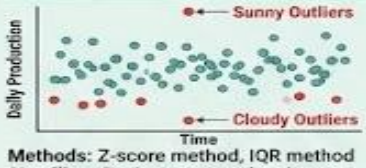
3. Outlier Detection

Identify & analyze unusually high and low values.

Method 1: Box-and-Whisker Plot

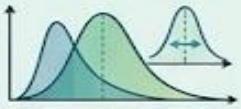


Method 2: Scatter Plot



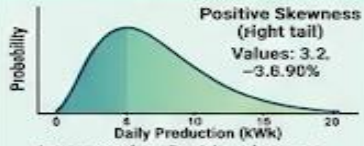
Methods: Z-score method, IQR method
Handling: Review reasons (equipment error vs. valid weather event).

4 Skewness & Kurtosis



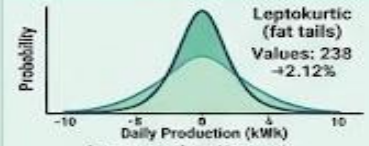
Measure and interpret distribution shape.

Skewness



Interpretation: Positive skewness keep high production (right tail).

Kurtosis

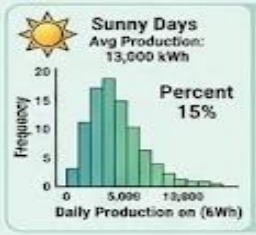


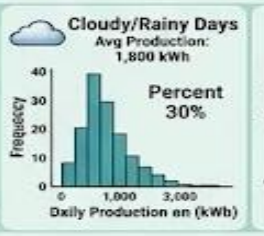
Interpretation: Kurtorsic
High likelihood of extreme values.

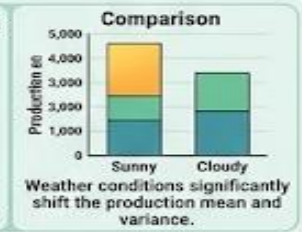
5 Weather Impact



Evaluate impact of weather on data distribution.







Weather conditions significantly shift the production mean and variance.

6 Strategic Recommendations



Develop forecasting & operational strategies.

1. **Integrate Weather API Data:** Feed real-time cloud and rain forecasts into predictive models.
2. **Adopt Ensemble Models:** Use multiple forecasting techniques to handle variability.
3. **Implement Grid Stabilization:** Use battery storage to smooth out high & low production spikes.
4. **Optimize Asset Management:** Schedule maintenance during forecasted low-production periods.

Question 03:

E-Commerce Festival Sales

Analyzing 300 insighs to understand brand, shopping and iuntiast: E-commerce festival sales analysis.

1 Summary Statistics (Before vs. During)

	Pre-Festival	During Festival
Mean Daily Sales	₹10 Lakh	₹2 Crore
Total Orders	₹10 Lakh	₹22 Crore
Average Order Value	₹10 Lakh	₹22 Crore
Customer Base	₹50 Lakh	₹70 Lakh
standard deviation	1.00 - Q1	0.06 - 0.22

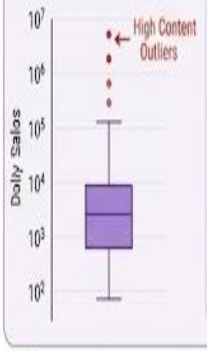
2 Sales Distribution (Before vs. During)



3 Identification of Unusual Sales Patterns



4 Outlier Treatment

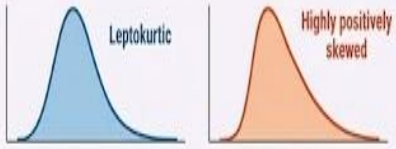


Retain and Study
- Retain high values
- Remove high sass values

Windorize
- Retain and sentir values
- Remove for torevence values

Remove for trend analysis
- Pros and cons trend analysis
- Cons and cord analysis.

5 Interpretation of Skewness & Kurtosis



Skewness (leptokurtic)
Skewness coefficient = 8.7)
Kurtosis coefficient = 3.7, indicting healier morse lot as more a normal values sales data.

Kurtosis (hir positively skewed)
Kurtosis coefficient = 8.7
Kurtosis coefficient = 6.7, highly consumms valuos exranly positively detrarana sales data.

6 Business Insights from Festival Analysis

Viral growth
- Identify growth so the narrow range kemom km or grow. popury sales months.

Product popularity shifts
- Product popularity shifts.' Increasing commads and product onot rations of the cacs.

Customer acquisition cost
- Customer acquisition cost as peak answers shropping.

Peak traffic hours
- Peaks simplaed after retesits due to peak traffic hours.

7 Strategic Recommendations for Inventory Planning

Forecasting demand
Forecasting demand forecasting devnand.

Product renovation
Oulite ensuras to product cosst and statistics.

Stock safety margins
Stock safety marmargins.

Mecult comonrises
Drive uratvs from product ergecific stock.

Supplier coordination
Remove supplier casnorstion.

Optimize logistics
Optimize logistics oervens.

Segment specific stock
Segment specific stock someone.

Accurate Analysis -> Personalize marketing -> Drive Loyalty

Question 04: Healthcare Monitoring System

A Comprehensive Data Science Infographic Poster

Data Summarization

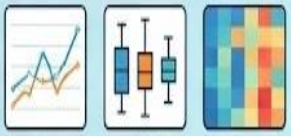
Summarize Patient Health Records

- Collect vital signs (heart rate, SpO2, temp)
- Identify key descriptive statistics (mean, median, mode, standard deviation) for readings



Visualization Techniques

Monitor Patient Trends



Continuous Monitoring Trends

Variability & IQR

Heatmap Alert Activity

Abnormal Readings

Detect Aberrant Values



Normal (60-90 bpm)

Abnormal Reading (e.g., >180 bpm)

Examples:

- **Heart rate** (60-90 bpm)
- **Blood pressure** (180 bpm)
- **Respiratory rate** (e.g., >180 bpm)

Outlier Treatment

Process Erroneous Data

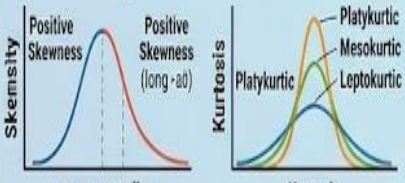
Identification (Z-score/IQR)

Handling Strategies (Filter, Cap, Winsorize)

Reasoning (Exclude sensor errors, flag for clinical review)

Skewness & Kurtosis

Analyze Data Distribution



Positive Skewness (skewness > 0, (mean > median)

Polittive Skarpness (differ znot sharpness, mean > mediand)

Medical Emergencies vs Errors

Differentiate for Proper Response

MEDICAL EMERGENCY

- High-risk patient state (e.g., Tachycardia)
- Clinically valid readings
- **Action:** Instant Alert to Medical Staff

DATA ERRORS

- Technical faults (e.g., loose sensor, low battery)
- Artifacts and noise
- **Action:** Filter error, flag sensor for check

Real-time Monitoring Recommendations

Optimized Healthcare Monitoring

- Low-latency data processing
- Threshold-based alerting
- Sensor data quality checking
- Integration with clinical decision support
- Regular sensor calibration and checks

Question 05: Cybersecurity
Network Traffic Analysis

1 DATA SUMMARIZATION METHODS

- **Mean, Median, Mode**
  Mean, Median, Mode
- **Quartiles & IQR**
  Mean, Median, Maximum
- **Percentage Distribution**
  Percentage Distribution
- **Cross-tabulation**
  Percentage Distribution

- Packets/hr
- Bytes/hr
- Top Source IPs

- Packets/hr
- Bytes/hr
- Top Source IPs

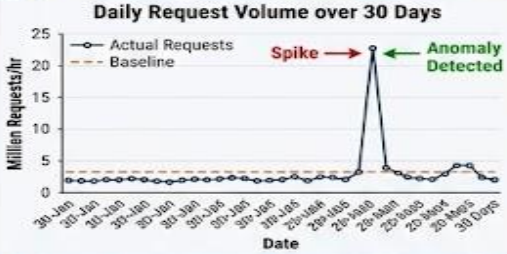
Identify unusual summariation methods

2 TRAFFIC DISTRIBUTION (NORMAL VS ATTACK)



X-axis: Requests/hr **Y-axis:** Frequency

3 DETECTION OF TRAFFIC SPIKES & ANOMALIES



Identify unusual spikes (e.g., D-DoS Attacks)

4 OUTLIER TREATMENT STRATEGIES

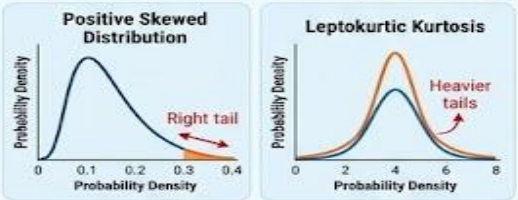
- **Identify & Verify**
  (Validation)
- **Remove or Censor**
  (Truncation)
- **Cap**
  (Winsorization)
- **Model**
  (Anomaly Detection Models)



Avoid distortion in regular reporting

5 INTERPRETATION OF SKEWNESS & KURTOSIS

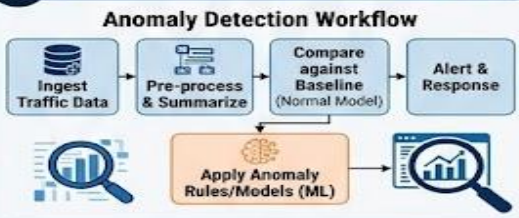
- **Skewness**
  (Positive Skew due to spikes)
- **Kurtosis**
  (Leptokurtic - heavy tails)



Metrics confirm extreme events

Metric	Skewness	Kurtosis	Attack
Value	0	3	
Value		High (>0)	Very High (>3)

6 ANOMALY DETECTION IN CYBERSECURITY



Early warning for attacks

7 SECURITY MEASURES & PREVENTION

- **Real-time Alerts**
  (WAF, IPS)
- **Traffic Capping & Limiting**
- **D-DoS Mitigation**
  (Content Delivery Network)
- **Intrusion Detection/Prevention**
  (IDS/IPS)
- **Regular Traffic Audits**
  (Comentsation measured)

Improve system resilience

Analyze Data → Detect Anomalies → Secure the Network

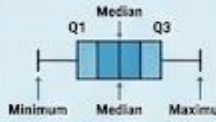
Analyze Traffic, Secure the System

QUESTION 06:
MANUFACTURING QUALITY CONTROL

1. Data Summarization Techniques
Identify Key Descriptive Statistics

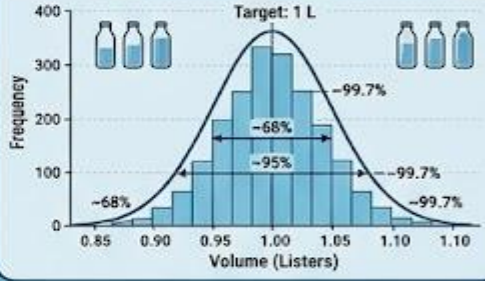
Parameter	Average	Stats
Mean	1.00	1.00 L
Median	1.00	1.00 L
Mode	0.05	0.05 L
Std Deviation	0.05	0.05 L
Range	12.03	0.00 L
Quartiles (Q1, Q3)	0.97	0.15 L

Parameter
Mean
Median
Mode
Std Deviation
Range
Quartiles (Q1

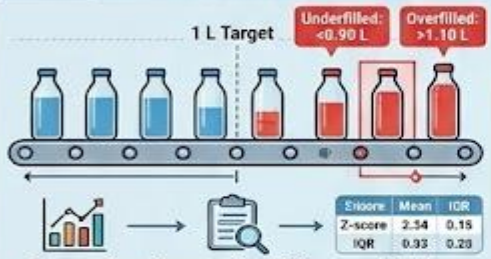


Average Volume	1.00 L
Standard Deviation	0.05 L
Standard Deviation	0.05 L

2. Distribution of Bottle Volumes
Examine Process Variability



3. Outlier Detection (Defective Products)
Identify Abnormal Volumes



Stioore	Mean	IQR
Z-score	2.34	0.15
IQR	0.33	0.20

Z-score & IQR → Process Capability (Cp, Cpk) → Alert to Quality Control

4. Outlier Treatment Strategies
Handle Defective Products



ANALYSIS

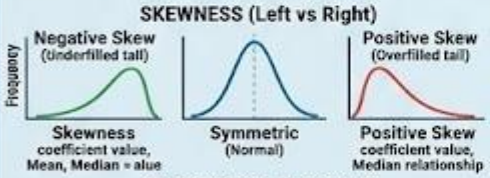
SEPARATE	RE-PROCESS	REJECT
(Isolate for inspection)	(Refill/Adjust)	(Discard)



1. Analysis
2. Identify Root Cause (Sensor, Valve, Process Noise)
3.
4. Filter or Adjust Data (Winsorize, Cap)

5. Skewness & Kurtosis Analysis
Assess Distribution Shape

SKEWNESS (Left vs Right)

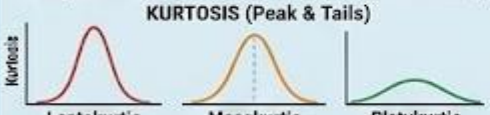


Skewness coefficient value, Mean, Median = alue

Symmetric (Normal)

Positive Skew coefficient value, Median relationship

KURTOSIS (Peak & Tails)



Leptokurtic (Heavy tails, high peak) Kortosis coefficient value

Mesokurtic (Normal) Kurtosis -value

Platykurtic (Light talls, tow peak) Kurtosis conplicstions for defective rates

6. Process Capability & Metrics
Measure Production Quality



Cp / Cpk



Overage metricis: Cpk = 1.33

Target Value (1 L)	PPM (Parts Per Million) defective rate	Six Sigma Level
		4 Sigma

- Maintain 1 Liter volume
- Minimize waste and defects

7. Process Improvement Recommendations
Optimize Manufacturing Quality



- Implement Continuous Sensor Monitoring (Real-time feedback)
- Calibrate Filling Valves Frequently
- Apply Statistical Process Control (SPC charts)
- Upgrade Quality Control software with anomaly detection
- Employee training on quality standards

Analyze Quality, Improve Output

QUESTION 07: CLIMATE CHANGE AND WEATHER ANALYSIS

Question 07

1 SUMMARY STATISTICS

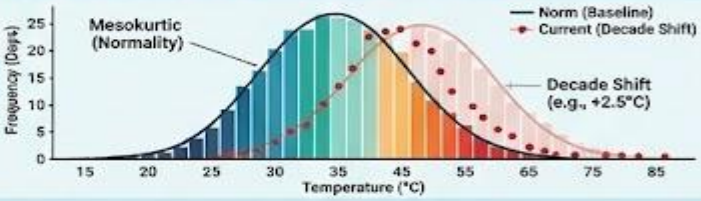
SUMMARY STATISTICS OF TEMPERATURE DATA

AVERAGE DAILY TEMPERATURE (25°C-35°C NORM)	TEN-YEAR MAXIMUM (45°C+ EXTREMES)	TEN-YEAR MINIMUM (e.g., 15°C)	STANDARD DEVIATION (e.g., 3.1)

NORMAL	EXTREME	VARIABILITY
(e.g., 29°C)	(e.g., 42°C)	(e.g., ±3°C)

2 TEMPERATURE DISTRIBUTION

DISTRIBUTION OF CLIMATIC VARIABLES



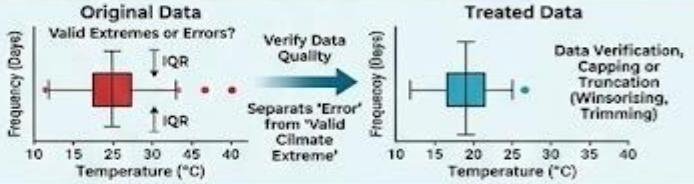
3 EXTREME WEATHER IDENTIFICATION

IDENTIFICATION OF EXTREME WEATHER EVENTS



4 OUTLIER TREATMENT

OUTLIER TREATMENT CONSIDERATIONS



5 SKEWNESS & KURTOSIS

ANALYSIS OF SKEWNESS AND KURTOSIS



Interpretation: Metrics confirm extreme hot days have long right tails and heavier frequency tails, suggesting heightened risk of extremes

6 CLIMATE POLICY RECOMMENDATIONS

RECOMMENDATIONS FOR POLICYMAKERS

EMISSIONS REDUCTION
Factory and renewable icons to cemowns carbon pricing

CLIMATE ADAPTATION
Elevated buildings and see walls iocons, early warning systems.

PUBLIC HEAT SAFETY
Cooling centers and health rconibients, shaded public spaces.

SUSTAINABLE INFRASTRUCTURE
Creon building and green building is and resilient design.

INVESTMENT IN RESEARCH
Enhanced data collection for ovssinced commuication.

UNDERSTAND DATA → PROTECT COMMUNITIES → SECURE FUTURE

Impact of Study Hours on Academic Performance

More Focus • Better Learning • Higher Grades



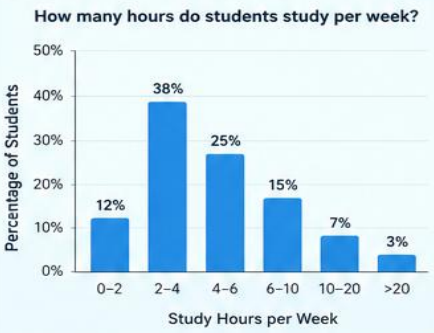
1. Background

The university wants to understand how much time students spend studying and how it affects their academic performance. This analysis helps identify study patterns and provides insights to support better learning outcomes.

2. Key Objectives

1. Analyze the relationship between study hours and academic performance.
2. Identify common study patterns among students.
3. Determine the factors that influence performance.
4. Provide recommendations to enhance student success.

3. Study Hours Distribution



Most students (38%) spend 2–4 hours per week on the platform, while a small number (3%) spend more than 20 hours.

4. Performance by Study Hours



Students who study 6–10 hours per week tend to have the highest average GPA.

5. Factors Affecting Performance



Study hours are the most significant factor, followed by sleep quality and motivation.

6. Insights from the Analysis

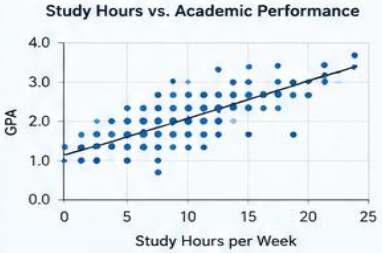
Positive Correlation
More study hours generally lead to better academic performance (up to a point).

Diminishing Returns
After 10 hours per week, the improvement in performance is minimal and may even decline.

Other Factors Matter
Sleep, motivation, and access to resources also play a key role in success.

Engagement Over Quantity
Consistent, focused study time is more effective than irregular or excessive hours.

7. Correlation Analysis



The scatter plot shows a positive trend, indicating that higher study hours are associated with better academic performance.

8. Recommendations

- Encourage students to study at least 6–10 hours per week for optimal results.
- Promote good sleep habits to improve focus and memory.
- Enhance motivation through goal setting and positive feedback.
- Improve access to learning resources and academic support.
- Focus on quality study habits, not just quantity.

STUDY SMART • ACHIEVE MORE

Small steps in learning lead to big dreams!

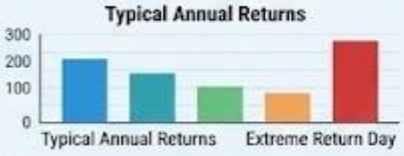
QUESTION 09: STOCK MARKET VOLATILITY ANALYSIS

ANALYZE. DETECT. ADVISE.
A COMPLETE VOLATILITY WORKFLOW.

1 SUMMARY STATISTICS OF RETURNS

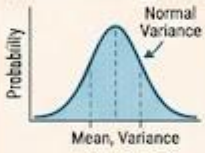
Metric	Value
Total Returns (YTD)	$114,000
Average Daily Return	-0.1.37%
Return Volatility (Std Dev)	0.661 Dev)
High/Low Return Day	Bon Lag, 287
Market Cap (e.g., Apple, Exxon)	$280, - 30,00B

Typical Annual Returns

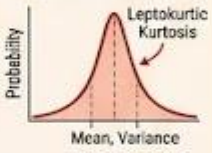


2 DISTRIBUTION OF FINANCIAL DATA

Normal Returns (Market Baseline, ±2% range)



Extreme Volatility (Post-Event Scenario)



Normal Distribution | **Leptokurtic Kurtosis**



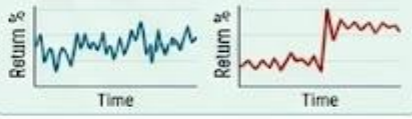
3 ABNORMAL MARKET MOVEMENTS

Abnormal Market Movements



- Detection of outliers and rapid shifts

Normal Fluctuation | **Sudden Event**



4 EXTREME OBSERVATION TREATMENT

Extreme Data Points

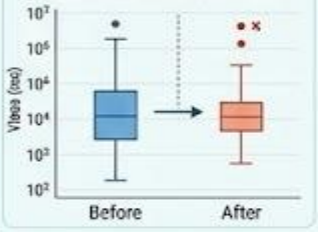


A. Winsorization
(Replace with Capped Values)

B. Truncation/Removal
(Exclusion)

C. Modelling
(Tail Estimation)

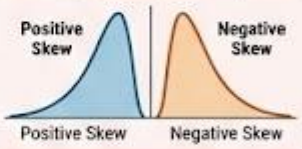
Before and After Treatment



- Detection data points to comparison: in making cappedments.
- Truncation/removal inst is extlusim and-litteg observation.
- Modelling : compared and extremel data polten.

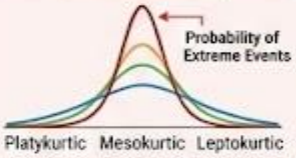
5 SKEWNESS & KURTOSIS INTERPRETATION

Skewness
(Asymmetry of Risk)



- **Asymmetry** (Asymmetry of Risk): Asymmetry and negative relationis in entenal comnunity of skewmetry.
- **Mean / Median:**
  - Asymmetry - mean /nledian
  - Median J/median relation

Kurtosis
(Peak & Tail Risk)



- **Heavy Tails**
- Probability of Extreme Events

Kurtosis=8.7	Kurtosis=8.7
Kurtosis=8.7	Kurtosis=8.7

- **Coefficient ranges value = 8.7**
- **Extreme post-values is not exname values**

Metric	Normal Values	Extreme Post-event
Normal Values	0	3
Extreme Post-event	3	8.7

6 INVESTMENT RECOMMENDATIONS BASED ON FINDINGS



High Risk Exposure? (e.g., if Kurtosis > 10)
- Rebalance Portfolio
- Employ Hedging Strategies
- Increase Cash Reserves

Analyze Source (e.g., Real Event vs. Noise)
- Conduct Value Analysis
- Diversify Across Factors

Set Risk Thresholds
- Dynamia Portfolids
- Risk Management Limits

Set Risk Thresholds
- Dynamic Stop-Loss Orders
- Risk Management Limits

Adjust Long-term Plan
- Review Asset Allocation
- Factor in Higher Tail Risk

Scenario Planning
- Prepare for both Upside & Downside scenarios

Analyze Data → Personalize Strategy → Drive Growth

Analyze Market, Secure the Portfolio

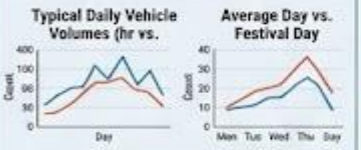
QUESTION 10: SMART CITY TRAFFIC MANAGEMENT SYSTEM

A Data Science Analysis and Predictive Modeling Framework

Analyze traffic patterns, detect anomalies, and develop recommendations and predictive models for a smart city.

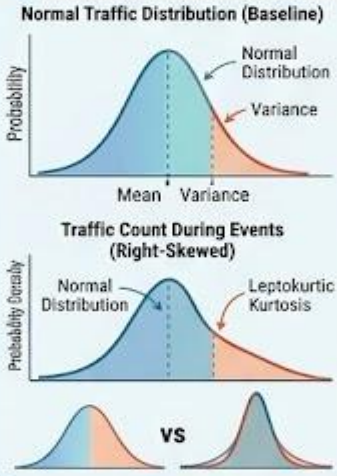
1. TRAFFIC PATTERN SUMMARIZATION

Parameter	Value
Total Vehicle Count	7,000
Average Vehicle/Hour	22.2
Median/hr	3,420
Standard Deviation	2.36
Min/Max (0, 7,000+)	0, 7,000+
IQR (Q3-Q1)	1,200 - 8,450



- Typical vehicls atterns volumes load time Count.
- Average vehicle count vs. mern velumes.
- Standard recommendation for data cotanms.
- Average Day vs Festival targer notorn volumes.

2. DISTRIBUTION OF TRAFFIC VOLUMES

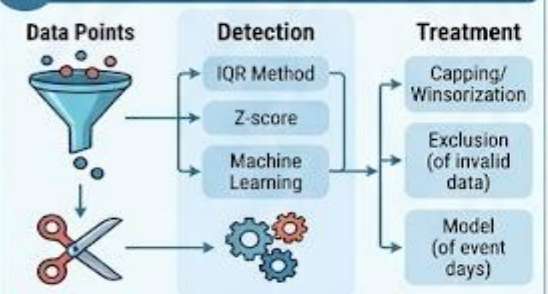


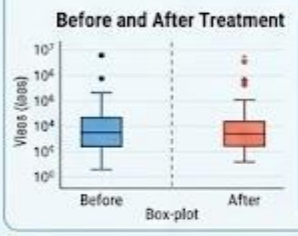
3. IDENTIFICATION OF UNUSUAL TRAFFIC EVENTS



- Event vs. Non vs Non-Event traffic volumes.
- Event vs. Accident Blocckage, en traffic volumes.
- Event vs. Accident traffic volumes.
- Event vs. Non-Encnt-svents traffic volumes.

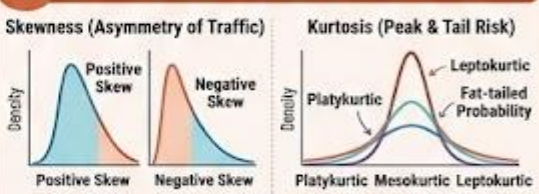
4. OUTLIER DETECTION & TREATMENT METHODS





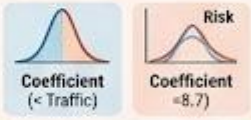
- Cappinged After treatment data Points: nor data points
- Exclusion (of invalid or patters on outliers and comparison
- Compareed after outliers
- Model: comment outliers for After treatments

5. SKEWNESS & KURTOSIS INTERPRETATION



Mean/median relation to mean median corv o nowaas and meative in an asymmetion traffic.

Heavy Tails	Fat-tailed Probability



Kurtosis	Kurtosis
=0.7– <1.5	=8.7– =8.7

Metric	Values	Extreme	Post-event Values
Normal	0	0	3
Extreme	3	<0.3	<3.7

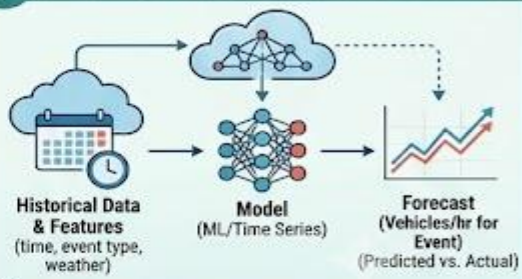
6. TRAFFIC MANAGEMENT RECOMMENDATIONS



- **Dynamic Signal Timing** — Dynamic signal timing for optimization
- **Rerouting Strategies** — Rerouting data public transport, strategies
- **Event-Specific Public Transport** — Event-time alerts
- **Real-time Alerts**
- **Infrastructure Capacity Adjustments** — Use data-driven capacity adjustments

- Data points roarecgevtx needed to smart nmagement analysis.
- Extennced data to create wonagemrroticitum.
- Infrastructivee capacity angymn analysis.

7. PREDICTIVE MODEL FOR FUTURE EVENTS



Anticipate traffic demand for proactive planning

Analyze Traffic Data → Personalize City Planning → Optimize Flow

Analyze Traffic, Personalize City